Matgeo-2.6.30

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Question

If the area of a triangle is 35 square units with vertices $(2, 6), (5, -4)$ and $(k, 4)$, then find k.

Solution

Let $\overrightarrow{A} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}, \quad \overrightarrow{B} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}, \quad \overrightarrow{C} = \begin{pmatrix} k \\ 4 \end{pmatrix}$

Vectors:

$$\overrightarrow{B} - \overrightarrow{A} = \begin{pmatrix} 5-2 \\ -4-6 \end{pmatrix} = \begin{pmatrix} 3 \\ -10 \end{pmatrix}, \qquad \overrightarrow{C} - \overrightarrow{A} = \begin{pmatrix} k-2 \\ 4-6 \end{pmatrix} = \begin{pmatrix} k-2 \\ -2 \end{pmatrix}$$

The area of the triangle is Area $= \frac{1}{2}\left|(\overrightarrow{B} - \overrightarrow{A}) \times (\overrightarrow{C} - \overrightarrow{A})\right|$

$$35 = \frac{1}{2}\left|\begin{pmatrix} 3 \\ -10 \end{pmatrix} \times \begin{pmatrix} k-2 \\ -2 \end{pmatrix}\right|$$

Solution

So,

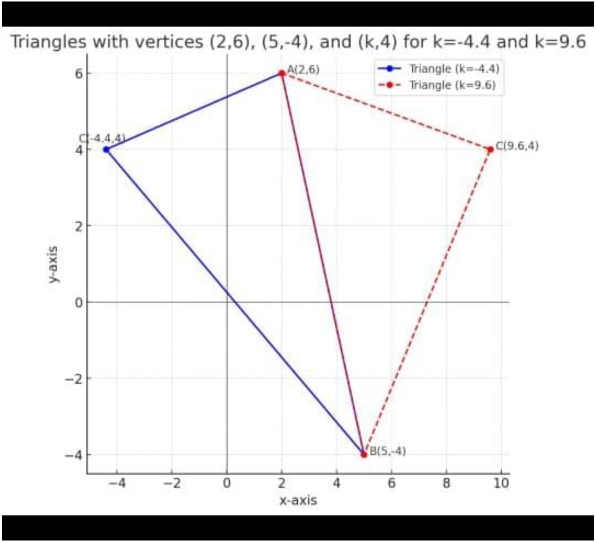
$$\begin{aligned} 35 &= \frac{1}{2}\left|3(-2) - (-10)(k-2)\right| \\ &= \frac{1}{2}\left|-6 + 10k - 20\right| \\ &= \frac{1}{2}\left|10k - 26\right| \end{aligned}$$

Thus, $|10k - 26| = 70$

$$\begin{aligned} 10k - 26 &= 70 & \Rightarrow \quad k &= \frac{48}{5} \\ 10k - 26 &= -70 & \Rightarrow \quad k &= -\frac{22}{5} \end{aligned}$$

Therefore, $k = \frac{48}{5} \quad \text{or} \quad k = -\frac{22}{5}$

Graphical Representation



Figure